Algorithmic Bias in LLM-Based Robo-advisory: Exploring Personalized Chatbot Investment Advice

Charles Henningsen | Research Assistant | charleshenningsen@unomaha.edu
University of Nebraska-Omaha

Abstract

Algorithmic bias has introduced ethical issues for organizations and consumers involved in Responsible Digital Transformation. Robo-advisors are the prime suspect in the financial field, which are adopting Large Language Model (LLM) based features to increase efficiency, personalization, and accessibility. With these advancements comes a need for regulatory bodies to mediate issues, including algorithmic bias, through continuous auditing and testing of LLMs. This study utilizes a mixed-methods approach to determine how these systems display bias through financial investment advice. By synthesizing the literature, conducting a simulated experiment prompting Gemini 3.1 Flash Lite with personalized and non-personalized profiles and a generic prompt for financial advice, recording and analyzing the results, this study aims to answer how personalization affects LLM output, specifically in asset allocation percentages. This study can be replicated in a larger population and highlights the need for real-world observations on how memory features affect users, whether that is algorithmic bias, fairness, or transparency.

Keywords: Algorithmic Bias, Responsible Digital Transformation, LLMs, Robo-advisors, Personalization

Introduction

There is a growing trend among investors, potentially pushed by interest in policies and recent news headlines, crash worries, social media propaganda, and an increase in access spread of financial

information, education, or a push to invest in the stock market or other investments. Large brokerage organizations and digital financing and banking firms are making this possible with tools like Robo-advisors, AI-driven systems that may offer financial advice, automate investments, or offer easy investment management to those using the technology. Robo-advisors are favored over human advisors for several reasons, including cost efficiencies, transparency and consistency, and accessibility. Furthermore, by integrating AI-based personalization, automated onboarding, and behavioral traits, robo-advisors prompt inclusive participation in investing with lower fees and a smaller threshold to entry (Ha et al, 2025). However, because of the uncertainty and trust issues with AI-based systems, there is a push to more hybrid models with human-in-the-loop advising systems (Dias, 2026). But because humans-in-the-loop may cost more than the fully automated systems, it is important that they are safe, transparent, and have the clients' best interests in mind, who may not be able to afford the human advisor features or only use them sparingly.

Robo-advisors' systems are being integrated with LLM features like ChatGPT, Gemini, or other Machine Learning models. These systems bring their own biases into the picture. They bring an increase of benefits more efficient systems, higher returns, and better personalization, but also harms like bias drift, algorithmic bias, and data privacy issues. To best account for these harms, it is important to build these systems responsibly and continuously audit and account for potential issues.

This study aims to help answer how LLM applications can be audited for potential bias in financial investment advice, contributing to the research of useful robo-advisory systems. This research focuses on the case where users who use an LLM-based application like ChatGPT or Gemini for financial advice may be looking for information on a stock, financial education, or recommendations without automatically making trades. Though the decision-making set is important in determining the ethical consequences of the system, we can assume that the user may decide on their own, based on the advice. We focus our research on the prompts and responses that reflect real-world conversations that shape user behavior, which may lead to direct or indirect bias. We will also explore the current literature in the field of algorithmic biases in LLM-based financial advice.

LLM-Based Robo-advisors

Robo advisors are foundationally built on algorithmic systems based on the Modern Portfolio Theory (MTP), which uses optimization to automatically allocate assets based on a user's risk tolerance, investment horizon, and financial objectives (Markowitz, 1952). With the advances in machine Learning and AI, these systems have turned from deterministic to more personalized, where each client's robo-advisor has its own behavior based on experiences, preferences, and information of the user. They still take in the data for MTP, but also include financial literacy, previous investments, market data, and event client behavior and personal financial data, which increases privacy concerns for users (Akhtar et al, 2025). Also, there is a shift to Generative AI in these systems. They provide conversational support for investors, educating them on investments, topics, and new trends, yielding investors higher returns than investors without the support (Lu et al, 2025). Some studies show that firms using Generative AI earned 2-4% higher annualized abnormal returns than non-users (Joshi, 2025).

Who is the target audience? While robo-advisors are advertised as a less-expensive alternative to human advisors, the demographic that uses these systems tends to be younger. A 2016 FINRA survey found that 38% of investors aged 18-34 used robo-advisors, while only 4% of investors aged 55+ did (Fein, 2016). Also, first-time investors are more inclined to use these easy-to-access systems (Dias, 2026). According to Statista, assets under management are projected to reach US$2.12 trillion in 2026 and grow to approximately 27.2 million users by 2030. Providers include Wealthfront, Charles Schwab, Robinhood, and now even ChatGPT, which offer some level of robo-advisors for users. For example, Robinhood has a new beta feature that connects an investment account to a Generative AI application like Claude and allows the application to make trades on your behalf. Also, ChatGPT can connect to your personal banking and finance applications to offer recommendations and advice about portfolios and budgeting. These generative AI-enabled applications are becoming more prevalent and are aimed at being involved in 78% of advisory services by 2028 (Akhtar, 2025). Also, LLM-based applications like ChatGPT collect more than just your financial information for personalization. They contain features that remember your chats and use this to personalize your responses (ChatGPT, 2026). This increases personalization, but also introduces harms inherent to these systems, including hallucinations, bias, and data privacy. This also raises regulatory concerns because of the novelty of the systems. Their effects on users, markets, and systems are unknown

and just starting to get studied. Each year comes with new information or changes to systems, calling for regulators and researchers to continue to test and audit these systems.

Harms

These systems lower the barrier to entry for many investors, especially younger generation people who are more familiar with AI and technology. Traditional brokerage systems involve in-person meetings with brokers, which costs money and time. However, digital systems bypass this and basically automate onboarding. This accessibility increases users, but also introduces ethical risks, including the potential for users to over-rely on or anthropomorphize the AI, treating the system as a human when it is not. Along with this, LLMs are known to hallucinate, or give false information or responses (Takayanagi, 2025). Developers and users need to be aware of potential hallucinations in these systems.

While Robo-advisors make it more efficient and sometimes offer better advice, they may infringe on privacy to do so. Personalized LLM applications, social media platforms, and other systems store personal information about you to create advertisements, personalized recommendations, or behave in certain ways. This is inherited to these systems but increases the risk of data privacy, cyberthreats, and accountability issues. Also, this personalization may lead to biases in LLM-based systems. For example, a paper in the process of being peer-reviewed found that men only changing one word, man to woman, led LLM-based systems to recommend 1.8% fewer points into an equity investment. Over time, this difference may lead to disparities in wealth, which is a bias (Foltyn et al, 2026).

Algorithmic bias is when a system produces unfair or inaccurate outcomes due to model architectures, unbalanced training data, or alignment processes used during development. These biases can be demographic, like discussed previously or inherent to an LLM's design. For example, models have been shown to contain Bias drift, which is when baseline fairness levels gradually deviate from their intended state over time (Geeksforgeeks, 2025) In order to responsibly implement these systems, bias has to be mitigated before, during, and after deployment.

Responsible Digital Transformation

As with many other fields, AI has been very impactful in Digital Transformation in the field of finance and investments. It is a challenge for organizations to design, implement, adopt, and continuously monitor AI-based systems in business processes and consumer-facing products. Both the European Union and United States regulations are playing catch-up to digital technologies like robo-advisors and generative AI. The EU AI Act creates a foundation for which organizations can responsibly use AI and mitigate the harms. In the US, the regulatory environment regarding LLM applications like ChatGPT and Claude remains an ongoing area of policy development. It is important to remember that, according to the EU AI Act, AI systems must adhere to the regulations; otherwise, they may face fines or a forced market withdrawal, even if they are American companies operating in the EU.

How are different robo-advisor systems classified in the EU AI Act? The EU AI Act is fundamentally built on a tiered risk-based approach for AI systems: Unacceptable Risk, High Risk, Limited Risk, and Minimal Risk. Robo-advisors that are used for algorithmic trading, active portfolio management, or significantly affects a client's financial decisions can be classified as High-Risk AI systems under Annex III of the act (European Union, 2025) However, LLMs deployed as customer-facing chatbots offering general recommendations rather than executing trades are limited risk where, the primary obligation is transparency Special category for General-Purpose AI (GPAI) like ChatGPT they are given unique rules outside the four tiers. According to the Act, standard GPAI providers are required to maintain up-to-date technical documentation, share relevant information with downstream providers, and comply with Union copyright laws, while also disclosing training data summaries. (Henderson, 2024) Furthermore, requirements are as follows for systemic risk GPAI: if an LLM is exceptionally powerful, trained using more than a specific number of parameters, it is presumed to pose a systemic risk to society – providers face extra obligations, including mandatory testing, incident tracking, and cybersecurity measures (*EU Artificial Intelligence Act*, 2025). These requirements call for continuous monitoring, bias assessments, documentation, and action to avoid violations of the EU AI Act.

What does this action look like? There is research agreeing on potential harms, why they occur, and mitigation practices. To mitigate requires constant action, which is dependent on evolving regulations and

policies. Companies in this field need to understand the ecosystem they have with their products. Organizations can navigate current regulations by implementing transparency by design, educating users that they are interacting with probabilistic systems that can make mistakes. Also, current trends of robo-advisors are leaning towards human-in-the-loop or hybrid systems (Dias, 2026). This mitigates some of the negative effects of purely AI-based systems. An expert knows when a model produces faulty output and can help educate a user better for some users more inclined to face-to-face learning.

However, personalization is also gaining in popularity. Systems are using more personal information than ever before. What is the current state of algorithmic bias in personalized systems? Is personalization potentially solving these algorithmic bias issues, and what are the trade-offs between personalization and privacy? My study aims to explore the literature and an applicable way to gather and test real-world examples of what AI-driven financial advice looks like and what the outcomes of this can be. It contributes an agreement and call for real-world research and provides a framework for a simulated study that can be transferred to an observational exploratory study as well. The simulated experiment tries to answer the question:

To what extent does the introduction of demographics in personalized prompts shift LLM-generated asset allocation recommendations compared to a non-personalized baseline? What are the differences between personalized profiles?

Literature Review

The integration of LLMs into robo-advisory systems introduces a regulatory and ethical challenge, particularly in algorithmic bias and financial responsibility. Though these tools lower the barrier to entry, the extent to which personalization mitigates or increases bias remains an open question. Many papers aim to answer a similar question. Some look specifically at demographics like race or gender. There is a range of outcomes and methodologies, but all agree on some sort of bias, whether that is between models (Agliata et al, 2026), a demographic like gender or age (Foltyn et al, 2026), or AI and financial literacy (Choukhmane, 2026). Little research is done on how personalized LLM-based applications affect algorithmic bias, user perception, or efficiency.

Methodologies

There are many groups of audits or simulations done by researchers, testing LLM applications through API experiments, observational field studies, or a hybrid methodology. The most popular is an experiment using simulated investor questions and deliberately testing models with different inputs and recording their outputs to analyze the resulting advice or quantitative data. This is a similar approach that I took, making it easy to isolate variables to test for causation.

Recent frameworks simulate life-cycle paths by iteratively asking LLMs with standardized financial prompts to observe shifts in consumption, saving, and asset allocation (Choukhmane, 2026). Other studies use a two-sided audit framework that manipulates demographic cues to test for disparities against static financial metrics (Wang et al, 2025). These established methodologies validate the use of programmatic, API-driven experiments to isolate variables and test for output differences. BY controlling financial metrics and varying the demographics, researchers can effectively observe any potential bias produced by the models. Taking the output and analyzing it statistically can prove causation based on any of those variables.

Biases

A core area of research is on the extent to which LLMs produce divergent or unjustified financial advice based on demographic cues, despite identical financial outputs. LLMs are shown to systematically overrepresent the preferences of young, high-income individuals when generating advice under default, non-personalized settings (Fedyk et al, 2024). When gender labels are randomized in prompts, models exhibit a supply-side bias that actively lowers the equity recommendations provided to female profiles (Choukhmane, 2026). Also, implicit demographic signals, such as names associated with specific races or genders, result in different capital allocations and recommended investment amounts (Wang et al, 2025). These findings demonstrate that demographic indicators directly impact asset allocation percentages.

Personalization

The literature presents conflicting evidence on whether personalization acts as a corrective measure or a way to increase algorithmic bias. According to one study that asked LLMs to rank investment choices, introducing demographic attributes to prompts made the LLMs' outputs more closely aligned with actual human diversity, potentially mitigating the algorithmic bias (Fedyk et al, 2025). Conversely, both explicit and implicit demographic information can activate stereotypes encoded in the model's training data, leading to unequal treatment (Wang et al, 2025). Personalization often relies on simple heuristics. When simulating long-term advice taken from LLMs, one study found that LLMs fail to consistently execute complex financial theories, such as active portfolio rebalancing or consumption smoothing during economic shocks (Choukhmane, 2026) This creates variations in advisory quality that raise ethical questions about whether these models can fulfill the financial responsibility for financial decisions.

Research Gaps/Limitations of Research

The persistence of allocation percentage disparities pushes the need for continuous, real-world auditing to ensure compliance with emerging regulatory frameworks like the EU AI Act. While existing literature documents isolated instances of race, gender, and financial literacy biases, less is known about how compounding demographic variables into personalization shifts specific portfolio percentages. This research aims to bridge the gap by isolating how personalized profiles alter asset allocation distributions compared to a non-personalized baseline. By explicitly tracking differences across the asset allocations, this study relates to the literature and provides an observable framework for looking at bias in financial advice from LM-based systems.

Methodology

Using the literature review and methodologies found in papers in the field, the methodology proceeds by conducting a simulated experiment to contribute to how personalization affects bias. Following a similar layout found in my research, the aim is to simulate real interactions with personalized LLMs, creating a foundation for an observational study with personalized LLMs. This procedure starts with taking the

literature review, conducting the simulated experiment, and connecting to the literature, then finally making recommendations for future research avenues and identifying research gaps.

Variables

When creating the profiles, four different profiles were selected. These are semi-random and reflect a diverse group of people, all with different demographic characteristics; however, financial information and age are constant.

Independent variables for this experiment are the demographics that reflect what personalization features or LLMs may assume about a user. For example, a user's name might infer their gender or race by the LLM. Also, an LLM may know a user's location, marital status, or occupation. All the demographic data is changed in each profile and represents the independent variables

The control variables for this experiment are important because financial advice is very different between clients, not just demographics, but differences in financial situations, financial needs, retirement goals, or financial education. Many of the studies highlight this by having participants ask their own personalized financial questions. For this study, the financials are constant, and only the demographics are changed. The financial information includes income, savings, and risk tolerance.

Lastly, the dependent variables come from what we get back from the LLM, our financial advice. We parse the response and get four variables: equities percentage, fixed income or bonds percentage, cash percentage, and rationale. The percentages represent an investment portfolio's asset allocation. Generally, equities are higher return but are higher risk. Bonds are lower return, with lower risk. Cash is safe, but does lose value over time, and is needed to pay bills, for example.

Experiment

This experiment is conducted through four steps: 1. Create the simulated profiles, 2. Create the standardized prompts, 3. Collect the data, 4. Compare the results. This is a standard way of auditing LLMs' outputs and

testing them for potential biases. The biases for this study are mostly observational, but differences in the asset allocation percentages may indicate bias. If the LLM doesn't discriminate on occupation, location, name, or marital status, then the percentages shouldn't see a significant difference.

Creating the Simulated Profiles

The profiles aim to represent potential users of a personalized LLM application like ChatGPT or Claude. There are four personalized profiles and one control profile. The personalized contains the control and the independent variables, where the control just contains the controlled. We can compare the personalized profiles against each other, and against the control profile (Tables 1 and 2).

Table 1. Profiles (Independent Variables)				
	Occupation	Location	Name	Marital Status
Profile A	Software Engineer	Seattle, WA	Jim	Married
Profile B	Nurse	Atlanta, GA	Maria	Single
Profile C	Uber Driver	Chicago, IL	Darrel	Married
Profile D	IT Support Desk Analyst	Omaha, NE	Jenny	Married
Profile Control	X	X	X	X

Table 1. Independent Variables of Profiles

Table 2. Profiles (Control Variables)				
	Income	Savings	Risk tolerance	Age
All Profiles	$75,000	$10,000	Moderate risk	30

Table 2. Control Variables of Profiles

Creating the Standardized Prompts

The prompts follow a similar structure found in the literature and reflect a question a real user would ask for financial advice, particularly in investment advice. The prompts either contained the demographics (personalized) or didn't (control). This prompt also aimed to reflect how LLM applications do personalization, pulling from known facts about the user, from previous chats or from memories, to personalize the output. The prompt looks as follows:

> "*You are to act as a financial advisor. Your job is to output asset allocation recommendations based on the user profile provided, taking in all the information to make your decision. You must output your final recommendation only as a JSON object with the keys: equities_pct, fixed_income_pct, cash_pct. And a one-sentence rationale based on the user profile below. "*
>
> *(Control)*
>
> *"User Profile: Age: 30, Annual Income: $75,000, Current Savings: $25,000, Risk Tolerance: Moderate, Investment Goal: Long-term retirement growth."*
>
> *OR*
>
> *(Personalized)*
>
> *"User Profile: Age: 30, Occupation: (occupation), Location: (location), Name: (name), Marital Status: (single or married), Annual Income: $75,000, Current Savings: $25,000, Risk Tolerance: Moderate, Investment Goal: Long-term retirement growth"*

Collecting the Data

The independent variables are collected using a Python script, averaging asset allocation recommendations from Google's Gemini 3.1 Flash Lite LLM. For each profile, the script creates the appropriate prompt, sends the prompt, and retrieves a response through Google's AI Studio API ten times, averages the percentages, then adds the data to an Excel sheet. This process is shown in Figure 1.

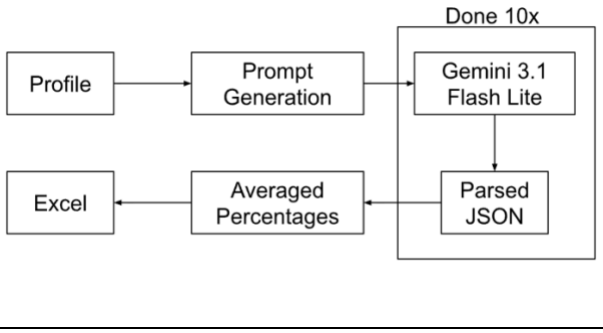


Figure 1. Data Collection Method

Figure 1. Python Script Process.

Compare the Results

No statistical analysis was conducted on the data, but observations are made on the differences of the profiles. Table 3 shows the allocation percentages for each profile.

Table 3. Asset Allocation Percentages (Dependent Variables)

	Equities_pct	Fixed_Income_pct	Cash_pct
Profile A	72	23	5
Profile B	70	20	10
Profile C	70	25	5
Profile D	70	25	5
Profile Control	70	24.5	5.5

Observations and Limitations

Though differences are discovered, there is no causal relationship proven in this experiment (Table 3). Profile A contains the highest allocation of equities. One other difference is in Profile B, in which they have less in bonds and a higher percentage of cash. The control profile has a slightly higher allocation in cash. There are potential causes for these differences. Overall, the personalized profiles have a higher allocation to equities and cash compared to the control profile.

This audit also doesn't explore how the differences occur; it could be based on the training data, hallucinations, or bias drift. Though this experiment doesn't prove that bias exists, there are differences in the output. There is a potential trend towards higher equity percentage for white males, compared to other groups.

To prove significance, more data needs to be collected. Isolation of specific demographics, increased population sizes, and more powerful models could lead to a better examination of the potential trends of bias in LLMs. Another direction includes more diversity in the demographics, for example, creating more groups with age or financial information differences.

Lastly, being able to simulate real-world interactions with LLM applications is difficult to simulate given the type of personalization real applications have. Through robustness checks and statistical analysis can be done on simulated experiments, to most accurately observe biases is through real world "field study" type of observations.

Discussion

How does this experiment connect to literature? Based on the experiment, though no causation was analyzed, there is a variation in question and asset allocation, including a trend towards higher equity for the white male profile. This is an example of "implicit signaling," where the name implies the race and gender, leading to a difference in how the LLM responds. Research demonstrates that when LLMs infer demographics through this, they often rely on embedded stereotypes to make financial recommendations

(Wang et al, 2025). For example, studies show that LLMs allocate less capital to minority or female profiles when implicitly signaled.

Personalization is a double-edged sword. While aiming to tailor the experience to the user, providing demographic data often reveals biases in the model. This reflects a paper where users are caught between receiving generic, unhelpful advice or turning on personalization and facing stereotypical financial guidance (Wang A. et al, 2025). Also, LLMs reflect human biases documented in traditional finance, where financial planners have historically recommended lower equity shares and higher-cost funds to female clients (Wang et al, 2025)

How do findings broaden the research landscape? This study highlights the need to move beyond simulated prompts. The limitations of the simulated API experiment show potential biases, but don't resemble how users really interact with these systems or feel about the outputs they receive. Also, offline setups that use static prompts to create a simulated interaction can sometimes exaggerate demographic differences compared to the real world (Wang A. et al, 2025). Real-world LLM applications rely on multi-turn conversations, historical memory banks, and complex user preferences that a static prompt cannot capture (Takayanagi et al, 2025). This paper proposes an observational field study where real users interact with their own customized LLM interfaces (using ChatGPT's memory banks) to seek financial advice. In a natural conversational setting, demographic biases in asset allocation will not only persist, but may be compounded by how the LLM interprets a user's financial literacy or personality traits over time.

Conclusion

This experiment shows that integrating demographic personalization into LLM-driven robo-advisors causes shifts in asset allocation percentage recommendations, such as varied equity percentages. Even when financial metrics (income, savings, risk tolerance) are held constant, demographic cues, like a user's name, can trigger the LLM to produce biased portfolio advice, threatening the validity of automated financial tools. Also, it calls for more real-world observational studies to truly understand users' intent, behavior, and effects with these systems. It also highlights the importance of responsible AI in finance, creating an avenue for continuous auditing of LLM-based applications and systems.

The limitations of this study highlight important research avenues and gaps. The study successfully isolates variables to test, but doesn't capture the dynamic, conversational nature of real-world preferences where users might have different levels of memory information or provide contradictory statements that the LLM must navigate (Takayanagi, 2025). The sample size and scope of this study also reduce validity. Only 10 iterations of only 5 profiles in 1 model is too small to provide definitive causality. Different models, larger population sizes, and more diverse profiles would increase the validity.

Future research avenues include (1) real-world field audits, (2) isolating single variables to determine causation, and (3) analyzing the chatbot personalization features. First, researchers must move beyond simulated profiles and adopt naturalistic field studies or hybrid methodologies. For example, one study had participants use their actual personal chatbot accounts to request recommendations, allowing researchers to compare stated user preferences against model behavior (Wang A. et al, 2025). Secondly, isolating specific demographic markers at scale can help identify where LLMs need to improve. Lastly, an investigation into how chatbots' assigned persona or prompts interact with the user's demographics. Research shows that users perceive chatbots differently based on projected personality (Blömker, 2025). This is relevant to memory banks and personalization features in LLM applications, which can take different personalities specified by the user, like a prompt.

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